From Parameter Estimation to Independent Evaluation

Weekly Problems in Number Theory, Probability, Statistics, and AI

November 2, 2026

This edition contains **seven main problems: two in number theory, one in probability, two in statistics, and two in AI**. It builds on the work you have reported completing. The new problems also revisit the issues identified in your recent checkpoints: probability weights, second moments, and standard errors.

The estimated times are **12, 12, 10, 16, 15, 40, and 45 minutes: about 150 minutes in total**, assuming you reuse your previously verified NumPy functions. Allow another 20–40 minutes if you build the training loops from scratch. Experiments with unequal class covariances are optional extensions for this edition.

All problems appear first, followed by the solutions. Keep an unassisted first attempt at each problem before using AI feedback.

Problems

1. Number theory: Removing specified prime factors [12 minutes]

Focus and new material: a finite prime-factor sieve, converting a pointwise restriction into a Dirichlet convolution.

Define

$$\chi(n) = \begin{cases} 0, & 2 \mid n, \\ (-1)^{(n-1)/2}, & 2 \nmid n, \end{cases} \qquad a = \mathbf{1} * \chi,$$

where

$$(f * g)(n) = \sum_{d \mid n} f(d) g(n/d), \qquad \mathbf{1}(n) = 1.$$

You may use the previously established identity

$$A(s) := \sum_{n \geq 1} \frac{a(n)}{n^s} = \zeta(s) L(s, \chi), \qquad \Re s > 1,$$

and the prime-power formulas

$$a(2^k) = 1, \quad a(p^k) = k + 1 \ (p \equiv 1 \bmod 4), \quad a(p^k) = \mathbf{1}_{\{k \text{ even}\}} \ (p \equiv 3 \bmod 4).$$

For a fixed positive integer q, let

$$a_q(n) = a(n) \mathbf{1}_{\{\gcd(n,q)=1\}}.$$

The product on the right is **pointwise multiplication**.

1. Find the Dirichlet series of a_q in the form $A(s)H_q(s)$, where H_q is a finite Euler product.

Proof.

$$\sum_{n\geqslant 1}\frac{a_q(n)}{n^s}=\prod_{p\mid q}\sum_{n\geqslant 0}\frac{a(p^n)}{p^{ns}}=\prod_{p}\sum_{n}\frac{a(p^n)}{p^{ns}}\left(\prod_{p\mid q}\sum_{n}\frac{a(p^n)}{p^{ns}}\right)^{-1}=A(s)H_q(s) \tag{1}$$

$$\implies H_q(s)\prod_{p\mid q}\sum_{n}\frac{a(p^n)}{p^{ns}}=1 \tag{2}$$

$$\implies H_q(s)\prod_{2\mid q}\sum_{n}\frac{1}{2^{ns}}\prod_{\substack{p\equiv 1 \pmod 4\\ p\mid q}}\sum_{n}\frac{n+1}{p^{ns}}\prod_{\substack{p\equiv -1 \pmod 4\\ p\mid q}}\sum_{n}\frac{1}{p^{2ns}}=1 \tag{3}$$

$$\implies H_q(s)=\prod_{2\mid q}\left(1-\frac{1}{2^s}\right)\prod_{\substack{p\equiv 1 \pmod 4\\ p\mid q}}\left(1-\frac{1}{p^s}\right)^2\prod_{\substack{p\equiv -1 \pmod 4\\ p\mid q}}\left(1-\frac{1}{p^{2s}}\right). \tag{4}$$

□

2. Construct a finitely supported multiplicative function h_q such that $a_q = h_q * a$. Give $h_q(p)$ and $h_q(p^2)$, treating $p = 2$ separately.

Proof.

$$H_q(s)\prod_{p\mid q}\sum_{n}\frac{a(p^n)}{p^{ns}}=1\implies\prod_{p\mid q}\left(\sum_{n}\frac{a(p^n)}{p^{ns}}\right)\left(\sum_{n}\frac{h_q(p^n)}{p^{ns}}\right)=1 \tag{5}$$

$$\implies\prod_{p\mid q}\sum_{n}\frac{(a*h_q)(p^n)}{p^{ns}}=1\implies h_q(n)=\begin{cases}a^{-1}(n)=(\mu*(\mu\chi))(n) & \forall p\mid n, p\mid q, n\leqslant q\\ 0 \qquad \text{otherwise}\end{cases}. \tag{6}$$

□

3. For $q = 10$, list every nonzero value of h_{10}. Explain why multiplication of the series and rearrangement of sums are valid.

Proof. $h_{10}(1)=1, h_{10}(2)=-1, h_{10}(5)=-2, h_{10}(10)=2$. h_q is a multiplicative function and only functions in $p \mid q$. □

Source and use: original extension, with the topic informed by Kiran Kedlaya, MIT 18.785, Spring 2007, "Dirichlet series and arithmetic functions," Sections 2–3, Exercise 4. The sieve construction is new to this worksheet.

2. Number theory: Local densities of weighted integers [12 minutes]

Focus and new material: a summatory main term with a coprimality restriction, and local densities under a finite cutoff.

Continue with Problem 1. Use the previously established estimate without reproving it:

$$S(x):=\sum_{n\leqslant x}a(n)=\frac{\pi}{4}x+O(\sqrt{x}).$$

1. For fixed q, prove
$$\sum_{\substack{n\leqslant x\\(n,q)=1}} a(n) = \frac{\pi}{4}H_q(1)x + O_q(\sqrt{x}).$$

Proof.
$$\sum_{\substack{n\leqslant x\\(n,q)=1}} a(n) = \sum_{n\leqslant x} a(n) \sum_{d\mid(n,q)} \mu(d) = \sum_{d\mid q} \sum_{dm\leqslant x} \mu(d)a(d)a(m) \tag{7}$$
$$= \sum_{d\mid q} \mu(d)a(d)S(x/d) = \frac{\pi}{4}x \sum_{d\mid q} \frac{\mu(d)a(d)}{d} + O(\sqrt{x}) = \frac{\pi}{4}x \cdot H_q(1) + O(\sqrt{x}). \tag{8}$$

□

2. Define a random integer N_x on $1, \ldots, \lfloor x \rfloor$ by
$$P(N_x = n) = \frac{a(n)}{S(x)}.$$

Find the limit of $P((N_x, 10) = 1)$, and compare it with the limit for a uniformly sampled integer.

Proof.
$$P((N_x, 10) = 1) = \frac{1}{S(x)} \sum_{\substack{n\leqslant x\\(n,10)=1}} a(n) \tag{9}$$
$$\implies \lim_{x\to\infty} P((N_x, 10) = 1) = H_{10}(1) = \frac{8}{25}. \tag{10}$$

□

3. Find the limits of $P(2 \mid N_x)$, $P(3 \mid N_x)$, and $P(5 \mid N_x)$. Prove that, for any **fixed finite collection of primes**, the corresponding divisibility indicators are independent in the limit.

Proof.
$$P(p \mid N_x) = \frac{1}{S(x)} \sum_{\substack{n\leqslant x\\p\mid n}} a(n) = \frac{1}{S(x)} \left(\sum_{n\leqslant x} a(n) - \sum_{\substack{n\leqslant x\\(n,p)=1}} a(n) \right) = 1 - \frac{1}{S(x)} \sum_{\substack{n\leqslant x\\(n,p)=1}} a(n) \tag{11}$$
$$\implies \lim_{x\to\infty} P(p \mid N_x) = 1 - H_p(1) \implies P(2 \mid N_x) = 1/2,\ P(3 \mid N_x) = 8/9,\ P(5 \mid N_x) = 16/25. \tag{12}$$

□

This problem weights integers in a finite interval by $a(n)$; it uses a different distribution from the earlier $a(n)n^{-s}$ construction.

Source and use: original design. The summation method is informed by Exercise 9 of Kedlaya's 2007 notes. The earlier proof of the hyperbola estimate is not repeated here.

3. Probability: Binomial moments by ordered counting [10 minutes]

Focus and new material: falling factorial moments; ordered versus unordered counting as a tool for risk calculations.

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$, with $n \geq 2$, $p \in [0,1]$, and $S = \sum_i X_i$. Define

$$(x)_r = x(x-1)\cdots(x-r+1), \qquad (x)_0 = 1.$$

1. Express $(S)_r$ as a sum of products indexed by distinct indices, and prove

$$\mathbb{E}[(S)_r] = (n)_r p^r.$$

Why are there $(n)_r$ terms rather than $\binom{n}{r}$? How should the identity be interpreted when $r > n$?

Proof. $(S)_r = \frac{S!}{r!(S-r)!} r! = \binom{S}{r} r!$. Thus, $(S)_r$ count the ways to arrange r variables X_i from S in total, considering order.
It is the cardinal of the set $\mathcal{I}_r = \{(i_1, \dots, i_r) \in \{1, \dots, n\}^r : i_a \neq i_b \quad (a \neq b)\}$.

$$\mathbb{E}\left[\sum_{\boldsymbol{i} \in \mathcal{I}_r} X_{i_1} X_{i_2} \cdots X_{i_r}\right] = \sum_{\boldsymbol{i} \in \mathcal{I}_r} \mathbb{E}[X_{i_1}]\mathbb{E}[X_{i_2}] \cdots \mathbb{E}[X_{i_r}] = p^r \sum_{\boldsymbol{i} \in \mathcal{I}_r} 1 = p^r (n)_r. \tag{13}$$

□

2. Use the following identities to find $\mathbb{E}[S^2]$, $\mathbb{E}[S^4]$, and $\mathbb{E}[((S)_2)^2]$:

$$S^2 = (S)_2 + S,$$
$$S^4 = (S)_4 + 6(S)_3 + 7(S)_2 + S,$$
$$((S)_2)^2 = (S)_4 + 4(S)_3 + 2(S)_2.$$

Proof.

$$\mathbb{E}[S^2] = \mathbb{E}[(S)_2] + \mathbb{E}S = (n)_2 p^2 + np = n^2 p^2 - np^2 + np. \tag{14}$$
$$\mathbb{E}[S^4] = \mathbb{E}[(S)_4] + 6\mathbb{E}[(S)_3] + 7\mathbb{E}[(S)_2] + \mathbb{E}[S] = (n)_4 p^4 + 6(n)_3 p^3 + 7(n)_2 p^2 + np. \tag{15}$$
$$\mathbb{E}[((S)_2)^2] = \mathbb{E}[(S)_4] + 4\mathbb{E}[(S)_3] + 2\mathbb{E}[(S)_2] = (n)_4 p^4 + 4(n)_3 p^3 + 2(n)_2 p^2. \tag{16}$$

□

Source and use: original design, connecting to Bernoulli sample estimation in MIT 18.650, Fall 2016, Problem Set 2, Problem 1. The falling factorial derivation is an original extension.

4. Statistics: Unbiased estimation versus maximum likelihood [16 minutes]

Focus and new material: exact mean squared error comparisons; distinguishing maximum likelihood, unbiasedness, and risk under squared loss.

Continue with Problem 3. To estimate p^2, compare

$$U = \frac{S(S-1)}{n(n-1)}, \qquad V = \left(\frac{S}{n}\right)^2.$$

The parameter space is $[0,1]$, so boundary MLEs are allowed for samples consisting entirely of zeros or entirely of ones.

1. Explain why V is the maximum likelihood estimator of p^2. Find the biases of U, V.

   Proof. $\mathrm{Bias}_{p^2} U = \mathbb{E}[U] - p^2 = 0, \mathrm{Bias}_{p^2} V = \frac{p}{n} - \frac{p^2}{n}$.

   $$L(p) = \prod_i f(X_i \mid p) = p^S (1-p)^{n-S} \tag{17}$$

   $$\hat{p}_{MLE} = \arg\max_p p^S (1-p)^{n-S} = \arg\max_p S \log p + (n-S)\log(1-p) = \frac{S}{n}. \tag{18}$$

   Thus, we conclude that $V = \hat{p}_{MLE}^2$ is the MLE of p^2. □

2. Use Problem 3 to obtain their exact MSEs. You may leave the answers in falling factorial form.

   Proof.

   $$\mathrm{MSE}_{p^2} U = \mathbb{E}[(U-p^2)^2] = \mathrm{Var}\, U = \frac{(n)_4 p^4 + 4(n)_3 p^3 + 2(n)_2 p^2}{n^2(1-n)^2} \tag{19}$$

   $$\mathrm{MSE}_{p^2} V = \mathbb{E}[(V-p^2)^2] = \mathbb{E}[V^2] - 2p^2\mathbb{E}[V] + p^4 \tag{20}$$

   $$= \frac{(n)_4 p^4 + 6(n)_3 p^3 + 7(n)_2 p^2 + np}{n^4} - 2p^2 \frac{n^2 p^2 - np^2 + np}{n^2} + p^4 \tag{21}$$

   □

3. Set $n = 2$. First tabulate the probabilities of $S = 0, 1, 2$ and the corresponding values of the two estimators. Compare their MSEs at $p = 0.1$ and $p = 0.5$.

   Proof.
   $$\begin{array}{lll} P(S=0) = (1-p)^2, & U = 0, & V = 0, \\ P(S=1) = p(1-p), & U = 0, & V = 1/4, \\ P(S=2) = p^2, & U = 1, & V = 1. \end{array} \quad \mathrm{MSE}\, U = p^2, \mathrm{MSE}\, V = \tfrac{p(1-p)(8p+1)}{8}$$
   □

4. For $n = 2$, find all $p \in (0, 1)$ for which V has lower risk. Does this contradict the earlier Rao-Blackwell improvement?

   Proof. For $p > \frac{\sqrt{33}-1}{16}$, we have $\mathrm{MSE}\, U > \mathrm{MSE}\, V$. It doesn't constitue contradiction to the ealier arguments. The $V \neq \mathbb{E}[U|S]$, we cannot deduce the inviality of Rao-Blackwell improvement. □

Source and use: adapted from the plug-in estimator bias problem in MIT 18.650, Fall 2016, PS2, Problem 1. The target is changed to p^2; the MSE comparison and boundary analysis are original extensions.

5. Statistics: What do zero test errors tell us? [15 minutes]

Focus and new material: the Wilson interval for a binomial proportion; applying standard errors and inversion of an acceptance region.

Training, validation, and model selection are complete. Fix the final classifier and denote its error probability on a new observation by r. An independent test set contains m identically distributed observations. Let K be the number of errors and $\hat{r} = K/m$.

1. Give the distribution of K, $\mathbb{E}[\hat{r}]$, and $\mathrm{Var}(\hat{r})$. Write down the estimated standard error of the sample mean.

Proof.

$$p(K=k)=\binom{m}{k}r^k(1-r)^{m-k} \Longrightarrow K\sim \text{Binomial}(m,r). \tag{22}$$

$$\mathbb{E}[\hat{r}]=\frac{1}{m}\mathbb{E}[K]=r, \quad \text{Var}\,\hat{r}=\frac{1}{m^2}\,\text{Var}\,K=\frac{r(1-r)}{m}. \tag{23}$$

□

2. Set $z=1.96$. Square and rearrange

$$\frac{|\hat{r}-r|}{\sqrt{r(1-r)/m}}\leqslant z$$

as a quadratic inequality in r, and obtain the Wilson interval. Handle endpoint cases using the rearranged inequality.

Proof. Given the inequality, we obtain

$$(\hat{r}-r)^2\leqslant z^2\frac{r(1-r)}{m} \tag{24}$$

$$\Longrightarrow \frac{2\hat{r}+\frac{z^2}{m}-\sqrt{4\hat{r}\frac{z^2}{m}-4\hat{r}^2\frac{z^2}{m}+(\frac{z^2}{m})^2}}{2+2\frac{z^2}{m}}\leqslant r\leqslant \frac{2\hat{r}+\frac{z^2}{m}+\sqrt{4\hat{r}\frac{z^2}{m}-4\hat{r}^2\frac{z^2}{m}+(\frac{z^2}{m})^2}}{2+2\frac{z^2}{m}}. \tag{25}$$

□

3. For $m=200, K=30$, calculate the standard error and Wilson interval. Then find the interval for $m=20, K=0$, and compare it with the ordinary Wald interval

$$\hat{r}\pm z\sqrt{\hat{r}(1-\hat{r})/m}.$$

Proof. $m=200, K=30$: SE $=\sqrt{\frac{\hat{r}(1-\hat{r})}{m}}=\frac{\sqrt{510}}{200}, 0.107135\leqslant r\leqslant 0.206057$.
$m=20, K=0$: SE $=0$. Wilson: $0\leqslant r\leqslant 0.16113$. Wald: $[0,0]$. □

4. Does this interval describe the new-observation error rate of the trained model, or variability from resampling the training set and retraining? Why does further model selection after inspecting test results undermine the assumptions of this inference?

Proof. For instinct, we assume that the Wilson interval describes the error rate of the trained model on its new-observation; we further prove this by stating that the Wilson interval is a confidence interval of the error rate r for a Fixed Final Classifier, giving the error probability on a new observation, base on the variance of the $\hat{r}$.
For the model selection base on the inspection of test results, it is equivalent to bring the test data in the training data, which destroys the independence of the test data. This way, the result cannot be proved to be effective statistically. □

The Wilson interval here uses a normal approximation; its finite-sample coverage is **not guaranteed to equal exactly** 95%.

Source and use: original application. The formula and inversion method are based on the NIST/SEMATECH Handbook, Section 7.2.4.1, which attributes the method to Wilson (1927). The data and training/test interpretation are new.

Shared experiment for Problems 6 and 7

The two AI problems share one classification experiment. The new task is to **estimate parameters from finite samples, train models, select hyperparameters, and evaluate on independent data**.

Use the generator below. Observations are independent, and the three datasets are mutually independent. Do not add standardization or PCA in this edition.

```
import numpy as np

def make_data(n, seed):
    rng = np.random.default_rng(seed)
    y = rng.binomial(1, 0.4, n)
    means = np.array([[-1., 0.], [1., 0.5]])
    cov = np.array([[1., 0.4], [0.4, 1.5]])
    X = rng.multivariate_normal(np.zeros(2), cov, n)
    return X + means[y], y

X_train, y_train = make_data(120, 20261102)
X_valid, y_valid = make_data(120, 20261103)
# Generate and use only after model selection in Problem 7:
# X_test, y_test = make_data(2000, 20261104)
```

The true parameters appear only in the generator. Fitting functions may receive only training data and the specified hyperparameters.

6. AI: Learning Gaussian discriminant analysis from labeled samples [40 minutes]

Focus and new material: joint likelihood estimation, bias and rank of a shared covariance estimate, regularization, and validation-based selection.

Context. Observe training data $X \in \mathbb{R}^{n\times d}$ and $y \in \{0,1\}^n$. Adopt the model

$$P(Y=1)=\pi, \qquad X_i \mid Y_i = k \sim N(\mu_k, \Sigma),$$

where $\mu_k \in \mathbb{R}^d$ and the shared covariance satisfies $\Sigma \succ 0$. All model parameters are unknown. Write $n_k = \#\{i : y_i = k\}$, initially assuming that both classes occur.

1. Write the joint log-likelihood and derive $\hat{\pi}, \hat{\mu}_0, \hat{\mu}_1$.

   Proof.

$$L(\pi,\mu_0,\mu_1) = \sum_{i:y_i=0} \log((1-\pi)N(\boldsymbol{x}_i;\mu_0,\Sigma)) + \sum_{i:y_i=1} \log(\pi N(\boldsymbol{x}_i;\mu_1,\Sigma)) \tag{26}$$

$$= n_0\log(1-\pi) + n_1\log\pi + \sum_{i:y_i=0}\log(N(\boldsymbol{x}_i;\mu_0,\Sigma)) + \sum_{i:y_i=1}\log(N(\boldsymbol{x}_i;\mu_1,\Sigma)). \tag{27}$$

$$(\hat{\pi},\hat{\mu}_0,\hat{\mu}_1) = \arg\max_{(\pi,\mu_0,\mu_1)} L(\pi,\mu_0,\mu_1) \Longrightarrow \hat{\pi} = \frac{n_1}{n}, \tag{28}$$

$$(\hat{\mu}_0, \hat{\mu}_1) = \arg\min_{(\mu_0,\mu_1)} \underbrace{\sum_{i:y_i=0} (\boldsymbol{x}_i - \mu_0)^\top \Sigma^{-1} (\boldsymbol{x}_i - \mu_0) + \sum_{i:y_i=1} (\boldsymbol{x}_i - \mu_1)^\top \Sigma^{-1} (\boldsymbol{x}_i - \mu_1)}_{F(\mu_0,\mu_1;X)}. \tag{29}$$

$$\nabla F(\mu_0, \mu_1) = 0 \iff \mu_0 = \frac{1}{n_0} \sum_{i:y_i=0} \boldsymbol{x}_i,\ \mu_1 = \frac{1}{n_1} \sum_{i:y_i=1} \boldsymbol{x}_i \tag{30}$$

$$\implies (\hat{\mu}_0, \hat{\mu}_1) = \left(\frac{1}{n_0} \sum_{i:y_i=0} \boldsymbol{x}_i, \frac{1}{n_1} \sum_{i:y_i=1} \boldsymbol{x}_i \right). \tag{31}$$

□

2. Let row i of the residual matrix R be $x_i - \hat{\mu}_{y_i}$. Derive

$$\hat{\Sigma} = \frac{1}{n} R^\top R.$$

You may use $\nabla_\Omega \log\det\Omega = \Omega^{-1}$, where $\Omega = \Sigma^{-1}$. Further prove that, conditional on the labels and with both classes present,

$$\mathbb{E}[\hat{\Sigma} \mid y] = \frac{n-2}{n}\Sigma.$$

Proof.

$$L(\Sigma) = \sum_{i:y_i=0} \log((1-\pi) N(\boldsymbol{x}_i; \mu_0, \Sigma)) + \sum_{i:y_i=1} \log(\pi N(\boldsymbol{x}_i; \mu_1, \Sigma)) \tag{32}$$

$$\hat{\Sigma} = \arg\max_{\Sigma} L(\Sigma) \tag{33}$$

$$\iff \hat{\Sigma} = \arg\min_{\Sigma} \underbrace{n \log(\det\Sigma) + \sum_{i \leqslant n} (\boldsymbol{x}_i - \hat{\mu}_{y_i})^\top \Sigma^{-1} (\boldsymbol{x}_i - \hat{\mu}_{y_i})}_{F(\Omega)} \tag{34}$$

$$\nabla F(\Omega) = -n\Sigma + \sum_{i \leqslant n} (\boldsymbol{x}_i - \hat{\mu}_{y_i})(\boldsymbol{x}_i - \hat{\mu}_{y_i})^\top = 0 \tag{35}$$

$$\implies \hat{\Sigma} = \frac{1}{n} \sum_{i \leqslant n} (\boldsymbol{x}_i - \hat{\mu}_{y_i})(\boldsymbol{x}_i - \hat{\mu}_{y_i})^\top = \frac{1}{n} R^\top R. \tag{36}$$

Now, we discuss whether the statistic $\hat{\Sigma}$ is an unbiased estimator of real covariance matrix Σ, while $\boldsymbol{y}$ is known.

$$\mathbb{E}[\hat{\Sigma} \mid \boldsymbol{y}] = \frac{1}{n} \sum_{i \leqslant n} \mathbb{E}[(\boldsymbol{x}_i - \hat{\mu}_{y_i})(\boldsymbol{x}_i - \hat{\mu}_{y_i})^\top \mid \boldsymbol{y}] \tag{37}$$

$$= \frac{1}{n} \sum_{i \leqslant n} \mathrm{Cov}(\boldsymbol{x}_i \mid \boldsymbol{y}) - \mathrm{Cov}(\boldsymbol{x}_i, \hat{\mu}_{y_i} \mid \boldsymbol{y}) - \mathrm{Cov}(\boldsymbol{x}_i, \hat{\mu}_{y_i} \mid \boldsymbol{y})^\top + \mathrm{Cov}(\hat{\mu}_{y_i} \mid \boldsymbol{y}) \tag{38}$$

$$= \frac{n+2}{n}\Sigma - \frac{1}{n} \sum_{i \leqslant n} \mathrm{Cov}(\boldsymbol{x}_i, \hat{\mu}_{y_i} \mid \boldsymbol{y}) + \mathrm{Cov}(\boldsymbol{x}_i, \hat{\mu}_{y_i} \mid \boldsymbol{y})^\top \tag{39}$$

$$= \frac{n+2}{n}\Sigma \tag{40}$$

$$-\frac{1}{n}\left(\sum_{i:y_i=0}\mathrm{Cov}(\boldsymbol{x}_i,\hat{\mu}_0\mid\boldsymbol{y})+\mathrm{Cov}(\boldsymbol{x}_i,\hat{\mu}_0\mid\boldsymbol{y})^\top+\sum_{i:y_i=1}\mathrm{Cov}(\boldsymbol{x}_i,\hat{\mu}_1\mid\boldsymbol{y})+\mathrm{Cov}(\boldsymbol{x}_i,\hat{\mu}_1\mid\boldsymbol{y})^\top\right) \quad (41)$$

$$=\frac{n+2}{n}\Sigma-\frac{1}{n}4\Sigma=\frac{n-2}{n}\Sigma \quad \text{(By the linearity of covariance).} \quad (42)$$

□

3. Prove $\mathrm{rank}(R)\leqslant n-2$. Discuss three cases: one class is missing, one class contains only one observation, and $n-2<d$. Explain what

$$\hat{\Sigma}_\gamma=\hat{\Sigma}+\gamma I,\qquad \gamma>0$$

can and cannot fix.

Proof. We recall that $R=\begin{pmatrix}(\boldsymbol{x}_1-\hat{\mu}_{y_1})^\top\\(\boldsymbol{x}_2-\hat{\mu}_{y_2})^\top\\\vdots\\(\boldsymbol{x}_n-\hat{\mu}_{y_n})^\top\end{pmatrix}$.

$$\mathrm{rank}(R)=\mathrm{rank}\left(\begin{pmatrix}\frac{1}{n_0}\left(\sum_{i:y_i=0}(\boldsymbol{x}_i-\hat{\mu}_0)\right)^\top\\\frac{1}{n_1}\left(\sum_{i:y_i=1}(\boldsymbol{x}_i-\hat{\mu}_1)\right)^\top\\\vdots\\(\boldsymbol{x}_n-\hat{\mu}_{y_n})^\top\end{pmatrix}\right)=\mathrm{rank}\left(\begin{pmatrix}0\\0\\\vdots\\(\boldsymbol{x}_n-\hat{\mu}_{y_n})^\top\end{pmatrix}\right)\leqslant n-2. \quad (43)$$

For the 3 special cases, we have

$$\begin{cases}\text{One class is missing: if we have only 1 class, we can only have 1 mean, } \mathrm{rank}(A)\leqslant n-1,\\ \text{One class contains only 1 observation: that class is reduced to 0, } \mathrm{rank}(A)\leqslant n-2,\\ n-2<d\text{ : we must have } \mathrm{rank}(A)\leqslant d.\end{cases}$$

The regularization of $\hat{\Sigma}_\gamma$ fix problems that $\mathrm{rank}\left(\hat{\Sigma}_\gamma\right)=n$, and $\hat{\Sigma}_\gamma$ has larger eigenvalues $\lambda+\gamma$. It is definite to derive the inverse matrix. Statistically, regularization makes bias. However, its MSE might be smaller. □

4. Implement `fit_gda(X, y, gamma) -> theta`, returning $\theta=(b,w)\in\mathbb{R}^{d+1}$ so that the predicted log-odds are $b+x^\top w$. Solve a linear system instead of explicitly forming an inverse. Check the following data:

```
X = np.array([[-2.,0.], [0.,0.], [0.,1.], [2.,1.]])
y = np.array([0,0,1,1])
```

For $\gamma=0.5$, obtain $\theta=(-1,4/3,2)$ to within 10^{-10}. For $\gamma=0$, detect the singularity.

5. On the main dataset, select from $\gamma\in\{0,0.01,0.1\}$ using validation **mean log loss**, breaking ties in favor of the larger γ. Use first the initial 30 training observations and then all 120. Record validation error and log loss. Keep the 120-observation model for final evaluation; do not inspect the test set yet.

6. Before running the experiment, predict the effect of increasing the sample size; afterward, explain your results. Why need validation loss not decrease in every single experiment? If the true class covariances differ, what limitation can remain in the shared-covariance model even with infinitely many observations?

Proof. If we increase the sample size n, then we expect the following effects on the statistics.

$$\hat{\pi} : n \uparrow \Longrightarrow \hat{\pi} \to \pi, \tag{44}$$

$$\hat{\mu}_0, \hat{\mu}_1 : n \uparrow \Longrightarrow \hat{\mu}_0 \to \mu_0,\ \hat{\mu}_1 \to \mu_1, \tag{45}$$

$$\hat{\Sigma}_\gamma : n \uparrow \Longrightarrow \mathbb{E}[\hat{\Sigma}_\gamma \mid \boldsymbol{y}] = \mathbb{E}[\hat{\Sigma} \mid \boldsymbol{y}] + \gamma I = \frac{n-2}{n}\Sigma + \gamma I \to \Sigma + \gamma I, \tag{46}$$

$$\text{Log Loss} : -\frac{1}{n}\sum_{i=1}^{n} y_i \log \hat{p}_i + (1-y_i)\log(1-\hat{p}_i),\ \left(\hat{p}_i = \frac{1}{1+e^{-\boldsymbol{w}^\top \boldsymbol{x}-b}}\right) \tag{47}$$

$$= \frac{1}{n}\sum_{i \leqslant n} \log(1+e^{\boldsymbol{w}^\top \boldsymbol{x}_i + b}) - y_i(\boldsymbol{w}^\top \boldsymbol{x}_i + b),\ \text{it converges when } n \to \infty. \tag{48}$$

$$\text{Classification Error} : \frac{1}{n}\sum_{i \leqslant n} \mathbf{1}_{\hat{\boldsymbol{y}}_i \neq \boldsymbol{y}_i},\ \text{it converges to a number not necessarily be 0.} \tag{49}$$

The result of the experiment is

Metric	$n = 30$	$n = 120$
optimal γ	0	0
Log Loss	0.6566533795873882	0.6581928537455757
Classification Error	0.175	0.38333333333333336

The randomness of the sample renders the Log loss and error non-monotonic. However, we can still distinguish the optimal γ. If $\Sigma \mid Y = 1 \neq \Sigma \mid Y = 0$, then we the log-odds is no longer linear $\boldsymbol{w}^\top \boldsymbol{x}_i + b$. Our model based on the linear log-odds can never fit the quadratic term. □

Source and use: the parameter derivation and fitting task are adapted from Stanford CS229, Summer 2019, PS1, Problem 1(d)–(e). Covariance bias, rank, failure conditions, and the experimental protocol are original extensions.

7. AI: Training logistic regression and separating optimization from generalization [45 minutes]

Focus and new material: a complete training protocol, gradient checking, comparison under equal computational budgets, and independent test evaluation.

Context. Use the same data as in Problem 6. Set

$$Z = [\mathbf{1}, X] \in \mathbb{R}^{n \times (d+1)}, \quad \theta = (b, w), \quad P = \operatorname{diag}(0, 1, \ldots, 1),$$

$$p_i = \sigma(z_i^\top \theta), \qquad J_\lambda(\theta) = \frac{1}{n}\sum_i \left[\log(1 + e^{z_i^\top \theta}) - y_i z_i^\top \theta\right] + \frac{\lambda}{2}\|w\|^2.$$

The intercept is unpenalized and $\lambda > 0$.

1. Derive the gradient and Hessian, and write a numerically stable expression for the loss. If $w \sim N(0, \tau^2 I)$ and the intercept is unpenalized, explain the relationship between MAP estimation and λ.

   Proof.

   $$\nabla_{\boldsymbol{w}} J_\lambda(\theta) = \frac{1}{n}\sum_i \left(\frac{1}{1 + e^{-\boldsymbol{w}^\top \boldsymbol{x}_i - b}} - y_i\right)\boldsymbol{x}_i + \lambda \boldsymbol{w} \tag{50}$$

   $$H = \nabla^2_{\boldsymbol{w}} J_\lambda(\theta) = \frac{1}{n}\sum_i \frac{d}{d\boldsymbol{w}}\left(\frac{1}{1 + e^{-\boldsymbol{w}^\top \boldsymbol{x}_i - b}} - y_i\right)\boldsymbol{x}_i + \lambda I \tag{51}$$

   $$= \frac{1}{n}\sum_i p_i(1 - p_i)\boldsymbol{x}_i \boldsymbol{x}_i^\top + \lambda I. \tag{52}$$

   For MAP, we use Bayesian analysis to derive the best-fit θ for the observed data $X, \boldsymbol{y}$.

   $$p(\theta \mid \boldsymbol{y}, X) \propto p(\boldsymbol{y} \mid \theta, X)p(\theta),\ \hat{\theta}_{MAP} = \arg\max_\theta p(\theta \mid \boldsymbol{y}, X) \tag{53}$$

   $$\hat{\theta}_{MAP} = \arg\max_\theta p(\theta \mid \boldsymbol{y}, X) = \arg\min_\theta \log\frac{1}{p(\boldsymbol{y} \mid \theta, X)} + \log\frac{1}{p(\theta)} \tag{54}$$

   $$= \arg\min_\theta \sum_i y_i \log(1 + e^{-\boldsymbol{z}_i^\top \theta}) + (1 - y_i)\log(1 + e^{\boldsymbol{z}_i^\top \theta}) + \frac{1}{2\tau^2}\boldsymbol{w}^\top \boldsymbol{w} \tag{55}$$

   $$\Longrightarrow \arg\min_\theta J_{1/n\tau^2}(\theta). \tag{56}$$

   For the stableness of the loss function, we just shift

   $$J_\lambda(\theta) = \frac{1}{n}\sum_i \log\left(e^{-y_i(\boldsymbol{w}^\top \boldsymbol{x}_i + b) - m_i} + e^{(1-y_i)(\boldsymbol{w}^\top \boldsymbol{x}_i + b) - m_i}\right) + m_i + \frac{\lambda}{2}\|\boldsymbol{w}\|^2, \tag{57}$$

   $$m_i = \max\left\{-y_i(\boldsymbol{w}^\top \boldsymbol{x}_i + b), (1 - y_i)(\boldsymbol{w}^\top \boldsymbol{x}_i + b)\right\}. \tag{58}$$

   □

2. Prove that the objective has a unique finite minimizer when both classes occur. Check the assumptions: if every label is 1, does regularizing only w still guarantee a finite minimizer?

   Proof. When $\#\{i : y_i = 0\}, \#\{i : y_i = 1\} \geqslant 0$, we have to prove that $\exists! \hat{\theta} = (\hat{b}, \hat{\boldsymbol{w}}) = \arg\min_\theta J_\lambda(\theta)$. $\forall \boldsymbol{v} = \begin{pmatrix} b \\ \boldsymbol{w} \end{pmatrix} \neq 0$,

   $$H_\theta = \nabla^2_\theta J_\lambda(\theta) = \frac{1}{n} Z^\top S Z + \lambda P, \quad (S = \mathrm{diag}(p_i(1 - p_i))). \tag{59}$$

   $$\boldsymbol{v}^\top H_\theta \boldsymbol{v} = \frac{1}{n}(Z\boldsymbol{v})^\top S(Z\boldsymbol{v}) + \lambda \boldsymbol{v}^\top P \boldsymbol{v} = \frac{1}{n}(b\mathbf{1} + X\boldsymbol{w})^\top S(b\mathbf{1} + X\boldsymbol{w}) + \lambda \boldsymbol{w}^\top \boldsymbol{w} \tag{60}$$

   $$= \frac{1}{n}\sum_i p_i(1 - p_i)\left|\boldsymbol{x}_i^\top \boldsymbol{w} + b\right|^2 + \lambda\|\boldsymbol{w}\|^2 > 0. \tag{61}$$

Thus, $H \succ 0$. If $\exists\hat{\theta}$, it is unique. Furthermore, we use the fact that $\#\{i : y_i = 0\}, \#\{i : y_i = 1\} \geqslant 0$,

$$J_\lambda(\theta) = \frac{1}{n}\sum_i \log\Big(e^{-y_i(\boldsymbol{w}^\top \boldsymbol{x}_i + b)} + e^{(1-y_i)(\boldsymbol{w}^\top \boldsymbol{x}_i + b)}\Big) + \frac{\lambda}{2}\|\boldsymbol{w}\|^2 \geqslant \left|\boldsymbol{w}^\top \boldsymbol{x}_k + b\right| \tag{62}$$

$$\Longrightarrow \lim_{\|\theta\|\to\infty} J_\lambda(\theta) = \infty \Longrightarrow J_\lambda(\theta) \text{ is Coercive.} \tag{63}$$

$$\Longrightarrow \exists S = \{\theta : \|\theta\| \leqslant \delta\}\, s.t. (\exists\hat{\theta}, \hat{\theta} \in \mathbb{R}^{d+1}) \iff (\exists\hat{\theta}, \hat{\theta} \in S). \tag{64}$$

Since $J_\lambda(\theta)$ is continuous, by Extreme Value Theorem, the local minimum exists in S. Thus, global minimum exists. □

3. Implement `loss_grad(theta, Z, y, lam)`. Use the first 8 training rows, $\theta = (0.2, -0.3, 0.4)$, $\lambda = 0.1$, and centered finite differences with step $h = 10^{-6}$. Verify

$$\frac{\|g - g_{\text{num}}\|_2}{\max(1, \|g\|_2 + \|g_{\text{num}}\|_2)} < 10^{-7}.$$

Also check that the loss and gradient remain finite for scores of ± 1000.

4. Compare full-batch gradient descent with SGD using batches of 20. At every SGD step, sample indices uniformly **with replacement**, independently of previous steps. Prove conditional unbiasedness of its gradient. Each candidate model processes **24,000 observations**: 200 GD steps or 1,200 SGD steps. Initialize both methods at zero and use

$$L = \frac{\|Z\|_2^2}{4n} + \lambda, \qquad \eta_{\text{GD}} = \frac{1}{L}, \quad \eta_t^{\text{SGD}} = \frac{1}{L(1 + t/200)}.$$

Prove that L is an upper bound on the gradient's Lipschitz constant. Record the full training objective after every 2,400 processed observations.

Proof. We optimize the θ.

$$\text{GD: } \theta_{t+1} = \theta_t - \frac{1}{L}\nabla_\theta J_\lambda(\theta_t) = \theta_t - \frac{1}{L}\left(\frac{1}{n}\sum_i (p_i - y_i)z_i + \lambda P\theta_t\right) \tag{65}$$

$$\text{SGD: } \theta_{t+1} = \theta_t - \frac{1}{L(1 + t/200)}\left(\frac{1}{n}\sum_{\substack{i\in B \\ |B|=20}} (p_i - y_i)z_i + \lambda P\theta_t\right) \tag{66}$$

In order to prove that L is an upper bound on the gradient's Lipschitz constant, we maintain that $\forall\theta_1, \theta_2$,

$$\frac{\|\nabla J_\lambda(\theta_1) - \nabla J_\lambda(\theta_2)\|_2}{\|\theta_1 - \theta_2\|_2} \leqslant L. \tag{67}$$

However,

$$\forall\theta, \boldsymbol{v}, \frac{\boldsymbol{v}^\top H_\theta \boldsymbol{v}}{\|\boldsymbol{v}\|^2} = \frac{(Z\boldsymbol{v})^\top S(Z\boldsymbol{v}) + n\lambda\boldsymbol{v}^\top P\boldsymbol{v}}{n\|\boldsymbol{v}\|^2} \leqslant \frac{\frac{1}{4}\|Z\boldsymbol{v}\|_2^2 + n\lambda\|\boldsymbol{v}\|^2}{n\|\boldsymbol{v}\|^2} \leqslant \frac{\|Z\|_2^2}{4n} + \lambda \tag{68}$$

$$\Longrightarrow \|H_\theta\|_2 \leqslant L \Longrightarrow \|\nabla J_\lambda(\theta + \boldsymbol{h}) - \nabla J_\lambda(\theta)\|_2 \leqslant \int_0^1 \|H_{\theta + t\boldsymbol{h}}\|_2 \|\boldsymbol{h}\|\, dt \leqslant L\|\boldsymbol{h}\|. \tag{69}$$

□

5. For each optimization method, choose from $\lambda \in \{0.001, 0.01, 0.1\}$ by validation log loss, breaking ties in favor of the larger value. Reset the random seed to `20261105` for every SGD candidate. Fix the three final models: GDA, logistic regression trained by full-batch GD, and logistic regression trained by SGD.

6. Only now generate the test set. Report each model's error rate, mean log loss, and Wilson interval from Problem 5. Explain:

   - Does a lower training objective necessarily imply a lower test error rate?
     For my experiment, lower mean log loss indicates lower test error rate.
   - GDA and logistic regression both produce logistic-form posteriors. Why can their finite-sample parameter estimates differ?
     Because GDA and logistic regression take different approach. GDA focus on the joint distribution of $p(X, Y)$, while logistic regression focus on $p(X|Y)$ (recall that $\boldsymbol{y}$ is known in logistic regression).
   - How strong a conclusion about algorithm performance is justified by a small error-rate difference on a single test set?
     In my experiment, SGD, GD performs better than GDA. However, we can conclude no further for larger sample size ($\gg$) or multiple tests.

Source and use: the basic loss and convexity questions draw on Stanford CS229, Summer 2019, PS1, Problem 1(a)–(b). That problem uses Newton's method; this adaptation uses regularized GD/SGD. The interface also draws on the Problem 1(b) starter code actually read in the user-specified 2018 repository. Gradient checking, budget comparison, and the test protocol are original extensions. Personal implementations in this third-party repository are not treated as official solutions.

Solutions and checks

The solutions follow the problem and subquestion numbering. Attempt the problems independently before checking your work.

1. Solution

(1) Removing every positive power of each $p \mid q$ removes its Euler factor:

$$\boxed{H_q(s) = \prod_{p \mid q} (1 - p^{-s})(1 - \chi(p)p^{-s})}.$$

Therefore

$$\sum_n \frac{a_q(n)}{n^s} = A(s)H_q(s), \qquad \Re s > 1.$$

(2) Set $h_q(1) = 1$. For $p \mid q$, put

$$h_q(p) = -(1 + \chi(p)), \quad h_q(p^2) = \chi(p), \quad h_q(p^k) = 0 \ (k \geq 3).$$

For $p \nmid q$, set the value at every positive power to zero, and extend multiplicatively. In particular, if $2 \mid q$, then

$$h_q(2) = -1, \qquad h_q(4) = 0.$$

Comparing the local generating functions prime by prime gives $h_q * a = a_q$.

(3) We have

$$H_{10}(s) = (1 - 2^{-s})(1 - 5^{-s})^2.$$

The nonzero values are

d	1	2	5	10	25	50
$h_{10}(d)$	1	-1	-2	2	1	-1

and every other value is zero.

Since $0 \leqslant a(n) \leq \tau(n)$, and

$$\sum_n \tau(n)n^{-\sigma} = \zeta(\sigma)^2 < \infty \quad (\sigma > 1),$$

the relevant series converge absolutely. Multiplication by a finite Dirichlet polynomial in particular causes no rearrangement difficulty.

Checkpoint. Why is the sieve factor for $p \equiv 3 \bmod 4$ equal to $1 - p^{-2s}$, rather than $1 - p^{-s}$? Because a vanishes at odd powers of this prime.

2. Solution

(1) Problem 1 gives the exact identity

$$\sum_{n \leqslant x} a_q(n) = \sum_d h_q(d)S(x/d).$$

For fixed q, the right side has only finitely many terms. Thus it equals

$$\frac{\pi x}{4}\sum_{d}\frac{h_q(d)}{d}+O\left(\sqrt{x}\sum_{d}\frac{|h_q(d)|}{\sqrt{d}}\right),$$

which gives the desired estimate.

(2) We obtain

$$\lim_{x\to\infty}P((N_x,q)=1)=H_q(1).$$

For $q=10$,

$$\boxed{H_{10}(1)=\frac{1}{2}\left(\frac{4}{5}\right)^2=\frac{8}{25}}.$$

For uniformly sampled integers, the limit is $\varphi(10)/10=2/5$. The difference comes from weighting by $a(n)$.

(3) Let ρ_p denote the limiting probability of not being divisible by p. Then

$$\rho_p=\begin{cases}1/2, & p=2,\\ (1-1/p)^2, & p\equiv 1 \bmod 4,\\ 1-1/p^2, & p\equiv 3 \bmod 4.\end{cases}$$

The corresponding divisibility probabilities are

$$\boxed{\frac{1}{2},\quad \frac{1}{9},\quad \frac{9}{25}}\qquad (p=2,3,5).$$

For any finite set of primes T,

$$\lim P(p\nmid N_x \text{ for all } p\in T)=H_{\prod_{p\in T}p}(1)=\prod_{p\in T}\rho_p.$$

Finite inclusion–exclusion now gives the product probability for every combination of divisibility and nondivisibility, proving limiting independence. No interchange of a limit with infinitely many events is involved.

Checkpoint. Verify

$$\lim P(2\mid N_x,\ 5\mid N_x)=\frac{1}{2}\cdot\frac{9}{25}=\frac{9}{50}.$$

3. Solution

(1) For a realized sample, $(S)_r$ counts ordered choices of r distinct successful positions:

$$(S)_r=\sum_{\substack{i_1,\dots,i_r\\ \text{pairwise distinct}}}X_{i_1}\cdots X_{i_r}.$$

There are $(n)_r=r!\binom{n}{r}$ terms, each with expectation p^r. Hence

$$\boxed{\mathbb{E}[(S)_r]=(n)_r p^r}.$$

For $r > n$, there are no such ordered choices and both sides are zero.

(2) Substitution gives

$$\mathbb{E}[S^2] = (n)_2 p^2 + np,$$

$$\boxed{\mathbb{E}[S^4] = (n)_4 p^4 + 6(n)_3 p^3 + 7(n)_2 p^2 + np},$$

$$\boxed{\mathbb{E}[((S)_2)^2] = (n)_4 p^4 + 4(n)_3 p^3 + 2(n)_2 p^2}.$$

A common error is replacing an ordered count with an unordered one without inserting the factor $r!$.

Checkpoint. For $n = 2, r = 2$, we have $(S)_2 = 2X_1X_2$, so the expectation is $2p^2$.

4. Solution

Write $R_U = \mathbb{E}[(U - p^2)^2]$ and $R_V = \mathbb{E}[(V - p^2)^2]$.

(1) The Bernoulli likelihood is maximized at $\hat{p} = S/n$. Since $p \mapsto p^2$ is one-to-one on the parameter space, $V = \hat{p}^2$ is the MLE of the target parameter. By Problem 3,

$$\mathbb{E}[U] = p^2, \qquad \mathbb{E}[V] = p^2 + \frac{p(1-p)}{n}.$$

Thus U is unbiased, whereas V has upward bias.

(2) Using second moments,

$$R_U = \frac{(n)_4 p^4 + 4(n)_3 p^3 + 2(n)_2 p^2}{[(n)_2]^2} - p^4,$$

which simplifies to

$$\boxed{R_U = \frac{4p^3(1-p)}{n} + \frac{2p^2(1-p)^2}{n(n-1)}}.$$

Let $M_4 = \mathbb{E}[S^4]$. Then

$$\boxed{R_V = \frac{M_4}{n^4} - 2p^2\left(p^2 + \frac{p(1-p)}{n}\right) + p^4}.$$

The quantity being compared is MSE; comparing variances alone would omit the squared bias.

(3) For $n = 2$:

S	Probability	U	V
0	$(1-p)^2$	0	0
1	$2p(1-p)$	0	1/4
2	p^2	1	1

It follows that

$$R_U = p^2 - p^4, \qquad R_V = p(1-p)\left(p + \frac{1}{8}\right).$$

p	R_U	R_V	Lower risk
0.1	0.0099	0.02025	U
0.5	0.1875	0.15625	V

(4) Since

$$R_V - R_U = p(1-p)\left(\frac{1}{8} - p^2\right),$$

we have

$$\boxed{R_V < R_U \iff p > \frac{1}{2\sqrt{2}}} \quad (0 < p < 1).$$

This does not contradict Rao–Blackwell: the earlier result improves X_1X_2 by replacing it with $U = \mathbb{E}[X_1X_2 \mid S]$. It does not guarantee that U outperforms every biased estimator.

Checkpoint. Explain why both statements can hold: "V is biased" and "V has smaller MSE for some p."

5. Solution

(1) Conditional on the model after training and selection,

$$K \sim \text{Binomial}(m, r), \quad \mathbb{E}[\hat{r}] = r, \quad \text{Var}(\hat{r}) = \frac{r(1-r)}{m}.$$

The estimated standard error is

$$\widehat{\text{SE}}(\hat{r}) = \sqrt{\frac{\hat{r}(1-\hat{r})}{m}}.$$

(2) Squaring and rearranging gives

$$(m + z^2)r^2 - (2m\hat{r} + z^2)r + m\hat{r}^2 \le 0.$$

Its two roots give the endpoints:

$$\boxed{\frac{\hat{r} + \frac{z^2}{2m} \;\pm\; z\sqrt{\frac{\hat{r}(1-\hat{r})}{m} + \frac{z^2}{4m^2}}}{1 + \frac{z^2}{m}}}.$$

This is a concrete example of constructing a confidence interval by inverting an acceptance region. See the NIST explanation of the method.

(3) For $m = 200, K = 30$,

$$\hat{r} = 0.15, \qquad \widehat{\text{SE}} \approx 0.02525,$$

$$\boxed{I_{\text{Wilson}} \approx [0.1071,\ 0.2061]}.$$

For $m = 20, K = 0$,

$$\boxed{I_{\text{Wilson}} = \left[0, \frac{1.96^2}{20 + 1.96^2}\right] \approx [0,\ 0.1611]}.$$

The Wald interval collapses to $[0, 0]$, failing to represent the uncertainty left by a finite test sample.

(4) The interval concerns the population error rate of the current fixed model. It does not directly quantify variation from resampling the training data and retraining. Further selection based on the test set makes the final model depend on that set, so the initial treatment of its error indicators as independent Bernoulli observations for a fixed model no longer applies directly.

Checkpoint. Correct predictions on all 20 test observations do not imply a population error rate of zero.

6. Solution

(1) Ignoring constants, the joint log-likelihood is

$$\ell = n_1 \log \pi + n_0 \log(1-\pi) - \frac{n}{2} \log\det \Sigma - \frac{1}{2} \sum_i (x_i - \mu_{y_i})^\top \Sigma^{-1} (x_i - \mu_{y_i}).$$

Differentiating with respect to the parameters gives

$$\boxed{\hat{\pi} = \frac{n_1}{n}, \qquad \hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i}.$$

(2) Fix the estimated class means and let $C = R^\top R$. The part depending on the precision matrix is

$$\frac{n}{2} \log\det \Omega - \frac{1}{2} \operatorname{tr}(\Omega C).$$

At a stationary point, $n\Omega^{-1} = C$. Thus, when $C \succ 0$,

$$\boxed{\hat{\Sigma} = C/n}.$$

Conditional on the labels, the expected centered residual sum of outer products in class k is $(n_k - 1)\Sigma$. Hence

$$\mathbb{E}[C \mid y] = [(n_0 - 1) + (n_1 - 1)]\Sigma = (n-2)\Sigma.$$

The ML covariance estimate is therefore generally biased. Replacing the denominator by $n-2$ gives an unbiased correction when that denominator is positive, but produces a different estimator from the MLE.

(3) The residual rows sum to zero within each class, yielding two independent linear relations between rows. Therefore

$$\operatorname{rank}(R) \le \min(d, n-2).$$

- If one class is absent, its mean is not identifiable from the data. Adding a diagonal term to the covariance cannot supply that missing mean.
- If one class has only one observation, its mean is defined but its within-class residual is zero. Whether the shared covariance is full rank also depends on the other observations.
- If $n - 2 < d$, the shared covariance estimate must be singular.

When the residual scatter matrix is singular, the positive-definite covariance model has no finite covariance MLE: shrinking variance along a direction with zero residual scatter makes the likelihood unbounded.

For $\gamma > 0$,

$$v^\top(\hat{\Sigma} + \gamma I)v = \frac{\|Rv\|^2}{n} + \gamma\|v\|^2 > 0$$

for every nonzero v. Thus regularization makes the matrix invertible; it also changes the estimator.

(4) Substituting the estimates into the posterior gives

$$w = \hat{\Sigma}_\gamma^{-1}(\hat{\mu}_1 - \hat{\mu}_0),$$

$$b = -\frac{1}{2}(\hat{\mu}_1 + \hat{\mu}_0)^\top w + \log\frac{\hat{\pi}}{1-\hat{\pi}}.$$

A reference implementation is:

```
def fit_gda(X, y, gamma):
    if gamma < 0 or not (np.any(y == 0) and np.any(y == 1)):
        raise ValueError("Both classes required; gamma must be >= 0")
    n, d = X.shape
    means = np.stack([X[y == k].mean(axis=0) for k in (0, 1)])
    R = X - means[y]
    cov = R.T @ R / n + gamma * np.eye(d)
    w = np.linalg.solve(cov, means[1] - means[0])
    pi = y.mean()
    b = -0.5 * (means[0] + means[1]) @ w + np.log(pi / (1-pi))
    return np.r_[b, w]
```

The unregularized covariance for the toy data is $\mathrm{diag}(1, 0)$. Adding $0.5I$ gives the requested parameter values.

(5) Running the specified sampling sequence gives:

Training size	Selected γ	Validation error	Validation log loss
30	0	0.1750	0.392471
120	0	0.1500	0.359385

These are results for this fixed sample, not a monotonic pattern that must hold for every random seed.

(6) More training data generally help reduce uncertainty in parameter estimation, but a single validation result also reflects sampling variability. If the true covariances differ, the true log-odds generally contain a quadratic term. A shared-covariance GDA model can therefore retain misspecification because of its linear boundary. More data can estimate the restricted model more precisely without changing what the model can represent.

Checkpoint. Passing the generator's true means or covariance to `fit_gda` bypasses the parameter-estimation skill this problem is intended to test.

7. Solution

(1) Let $p = \sigma(Z\theta)$. Then

$$\boxed{\nabla J_\lambda = \frac{1}{n}Z^\top(p - y) + \lambda P\theta},$$

$$\boxed{\nabla^2 J_\lambda = \frac{1}{n} Z^\top D Z + \lambda P}, \qquad D_{ii} = p_i(1 - p_i).$$

For binary labels, the single-observation loss can be computed stably as

$$\ell(z, y) = \log\Big(1 + e^{(1-2y)z}\Big).$$

Use `np.logaddexp` instead of evaluating a large exponential directly.

The negative log objective for MAP is

$$\sum_i \ell_i + \frac{\|w\|^2}{2\tau^2}.$$

Dividing by n yields

$$\boxed{\lambda = \frac{1}{n\tau^2}}.$$

(2) For any nonzero direction $v = (v_b, v_w)$,

$$v^\top H v = \frac{1}{n} \sum_i p_i(1 - p_i)(z_i^\top v)^2 + \lambda \|v_w\|^2 > 0.$$

If $v_w \neq 0$, the regularization term is positive. If $v_w = 0$, then $v_b \neq 0$, and the first term is positive. The objective is therefore strictly convex.

Existence requires a separate argument. If the objective stays bounded, the regularizer keeps w bounded. When both classes occur, $b \to +\infty$ makes the negative-class loss diverge, and $b \to -\infty$ makes the positive-class loss diverge. Thus finite sublevel sets are bounded, and a unique minimizer exists.

If all labels are 1, take $w = 0, b \to +\infty$. The objective approaches zero but never attains it at a finite b. Penalizing only w does not guarantee existence.

(3) The core implementation is:

```
def augment(X):
    return np.column_stack([np.ones(len(X)), X])

def loss_grad(theta, Z, y, lam):
    scores = Z @ theta
    p = np.exp(-np.logaddexp(0., -scores))
    reg = theta.copy()
    reg[0] = 0.
    loss = np.logaddexp(0., (1-2*y)*scores).mean()
    loss += 0.5 * lam * (reg @ reg)
    grad = Z.T @ (p-y) / len(y) + lam * reg
    return loss, grad

def metrics(theta, X, y):
    scores = augment(X) @ theta
    error = np.mean((scores >= 0) != y)
    nll = np.logaddexp(0., (1-2*y)*scores).mean()
    return error, nll
```

Component j of the numerical gradient is

$$(g_{\mathrm{num}})_j = \frac{J(\theta + he_j) - J(\theta - he_j)}{2h}.$$

The specified check gave a relative error of approximately 7.8×10^{-11}. This is a deterministic numerical check, not Monte Carlo error.

(4) Conditional on the training data and current parameter, each sampled index is uniform. Hence

$$E\left[\frac{1}{B}\sum_{j=1}^{B} z_{I_j}\left(\sigma(z_{I_j}^{\top}\theta) - y_{I_j}\right) + \lambda P\theta \;\middle|\; \mathcal{D}, \theta\right] = \nabla J_\lambda(\theta).$$

Divide the data term by the **batch size** B and add the regularization gradient once.

Because $0 < p_i(1 - p_i) \le 1/4$,

$$H \preceq \frac{Z^{\top}Z}{4n} + \lambda I,$$

so the stated L is a valid upper bound. Full-batch GD with step size $1/L$ makes the training objective nonincreasing. Individual SGD steps need not do so.

```
def train_lr(X, y, lam, mode):
    Z = augment(X)
    n = len(y)
    L = np.linalg.norm(Z, 2)**2 / (4*n) + lam
    theta = np.zeros(Z.shape[1])
    rng = np.random.default_rng(20261105)
    B = n if mode == "GD" else 20
    history = []

    for t in range(24000 // B):
        idx = np.arange(n) if mode == "GD" else rng.integers(n, size=B)
        eta = 1/L if mode == "GD" else 1/(L*(1+t/200))
        theta -= eta * loss_grad(theta, Z[idx], y[idx], lam)[1]
        if ((t+1)*B) % 2400 == 0:
            history.append(((t+1)*B, loss_grad(theta, Z, y, lam)[0]))
    return theta, history
```

The extra diagnostic cost of recording the full objective is the same for both methods. The comparison counts processed observations, not iterations.

(5) For each method, evaluate the candidate λ values separately and select by unpenalized validation mean log loss. Both methods select $\lambda = 0.001$ in this run.

Method	Validation log loss
Full-batch GD	0.364026
SGD	0.364262

At this budget, full-batch GD is closer to the minimum of the regularized training objective. That does not determine the test ranking in advance.

(6) After completing all selection, generate the prescribed test set once. Reference results are:

Model	Hyperparameter	Errors/2000	Error rate	Log loss	Wilson interval
GDA	$\gamma = 0$	306	0.1530	0.345202	[0.1379, 0.1694]
LR, GD	$\lambda = 0.001$	304	0.1520	0.344893	[0.1369, 0.1684]
LR, SGD	$\lambda = 0.001$	302	0.1510	0.344233	[0.1360, 0.1674]

These values were verified by running the experiment. Use them to check sampling order, loss definitions, and implementation. Other random experiments need not reproduce this ranking.

Keep the following sources of differences separate:

- **Parameter estimation:** finite training data make parameter estimates differ from their population-optimal values.
- **Optimization:** how well GD or SGD minimizes the specified training objective.
- **Model assumptions:** GDA fits the joint distribution $p(x, y)$, whereas logistic regression fits the conditional distribution $p(y \mid x)$. Identical posterior forms do not imply identical finite-sample estimates.
- **Evaluation randomness and noise:** test error is a sample estimate of population error. Even with known true parameters, overlapping class distributions create irreducible classification error.

Differences of only 2–4 errors here do not establish that SGD generalizes better. The models' errors are paired because they use the same test observations. A formal comparison should use that pairing; overlap or nonoverlap of their individual intervals is not itself a formal test of the difference.

Checkpoint. If full-batch GD achieves a lower training objective while SGD makes two fewer test errors, explain both observations without treating them as contradictory.

Progress record for this edition

- **New capabilities provided:** finite prime-factor sieves and weighted densities; falling factorial moments; crossing risks of unbiased and maximum likelihood estimators; Wilson intervals; fitting GDA from training samples; and gradient checking, budget comparisons, and independent evaluation for logistic regression.
- **Necessary review:** ordered counting, probability normalization, second moment minus squared mean, standard errors, and conditional expectation given training data. Each supports a new problem in this edition.
- **Suggested next step:** after completing this edition, extend the same project to model misspecification and probability calibration, then proceed to Softmax. Repeated experiments with unequal covariances can take another 20–30 minutes or be left for a later edition. This edition has been *provided*; its completion has not yet been reported.